

Two-Phase PageRank with Travel-Time Self-Loops

Traffic Prediction on the Salt Lake City Road Network

Hamid

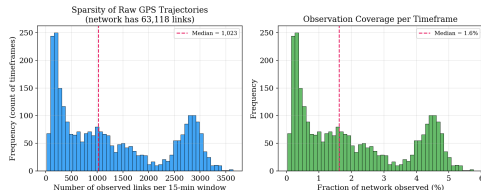
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Outline

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- 2 Building the Transition Matrix M_{2N}
- 3 Building the Teleportation Vector E_N
- 4 The Power Iteration
- 5 Smoothing Justification
- 6 Results
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The Traffic Prediction Problem

- Salt Lake City: **99,716 directed road links**
- GPS traces cover only a **tiny fraction** of the network per 15-min window
- Goal: predict the **full traffic distribution** across every road

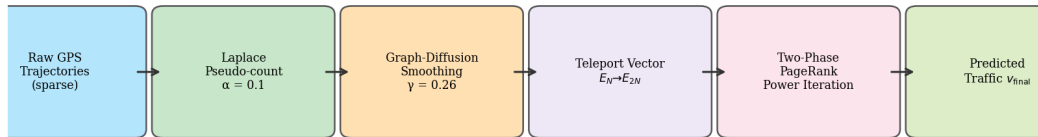


Our Approach

Model traffic as a **two-phase random walk** on the road graph, incorporating physical travel time

End-to-End Pipeline

Two-Phase PageRank: End-to-End Data Pipeline



Two Components of the Model

$$\mathbf{v}^{(k+1)} = \underbrace{d \cdot M_{2N}^\top \cdot \mathbf{v}^{(k)}}_{\text{Transition (topology)}} + \underbrace{(1 - d) \cdot E_{2N}}_{\text{Teleportation (ground truth)}}$$

Component	Built from	Role
Transition M_{2N}	Road topology, lengths, speeds	Where vehicles move
Teleportation E_{2N}	Ground-truth GPS counts	Where vehicles are born

The Two Phases of Driving

Every road link i has **two states**:

Up Phase

Actively traversing link i

Stays proportional to travel time

Self-loop traps mass

$\xrightarrow{\beta}$

Down Phase

Locally dispersing

Moves to adjacent links

Turning, parking, etc.

$\beta = 0.90$: drop probability from Up to Down per step

Up Phase: Travel-Time Self-Loops

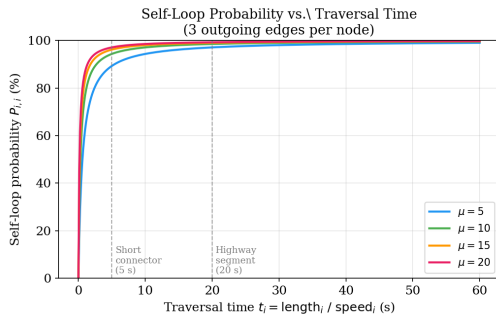
A vehicle on link i has traversal time $t_i = \ell_i / s_i$ seconds.

We add a **self-loop** so the vehicle “dwells” on the link proportionally:

$$w_i = \mu \cdot \frac{\ell_i}{s_i}$$

- ℓ_i = road length, s_i = speed limit
- $\mu = 20$ = dwell-time scale

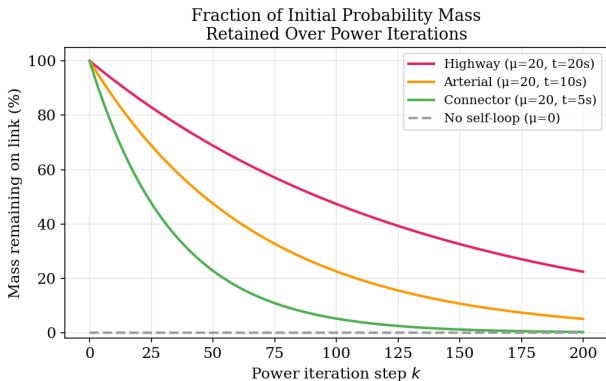
Longer/slower roads retain more mass



Self-Loop Probability

Row-stochastic transition after adding self-loop:

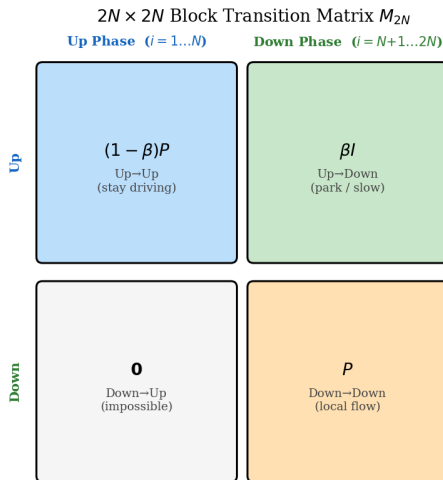
$$P_{\text{self}}(i, i) = \frac{w_i}{w_i + \deg^+(i)}, \quad P_{\text{self}}(i, j) = \frac{A_{ij}}{w_i + \deg^+(i)}$$



The $2N \times 2N$ Block Matrix

$$M_{2N} = \begin{pmatrix} (1-\beta) P_{\text{self}} & \beta I \\ 0 & P_{\text{adj}} \end{pmatrix}$$

- **Top-left:** Up→Up
(stay driving, with self-loops)
- **Top-right:** Up→Down
(drop to local dispersal)
- **Bottom-right:** Down→Down
(move to adjacent links)
- **Bottom-left:** 0
(no return from Down to Up)



Key Fact: M_{2N} Uses No Ground Truth

M_{2N} is built **entirely** from the **road network**: topology, lengths, and speed limits.
No GPS data. No traffic counts.

The transition matrix defines the “physics”—
the rules of how vehicles move through the network.

It is **built once** for a city and **reused** for every 15-minute prediction window.

Four-Step Construction of E_N

- ① **Extract raw GPS counts** $\mathbf{c} \in \mathbb{Z}_{\geq 0}^N$ (most $c_i = 0$)
- ② **Laplace pseudo-count:** $\tilde{c}_i = c_i + 0.1$ (eliminate zeros)
- ③ **Graph-diffusion smoothing** ($\gamma = 0.26$):

$$\mathbf{c}_{\text{smooth}} = 0.74 \tilde{\mathbf{c}} + 0.26 \mathbf{P}_{\text{adj}} \tilde{\mathbf{c}}$$

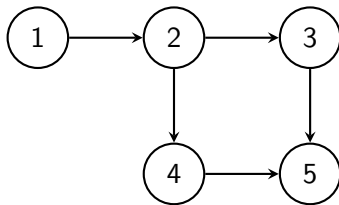
- ④ **L^1 -normalise:**

$$E_N[i] = \frac{c_{\text{smooth},i}}{\sum_j c_{\text{smooth},j}}, \quad \sum_i E_N[i] = 1$$

Result

$E_N[i]$ = probability that a new vehicle appears on link i

Worked Example: 5-Link Network



	Link 1	Link 2	Link 3	Link 4	Link 5
Raw count c_i	0	5	0	3	0
+ Laplace $\tilde{c}_i$	0.10	5.10	0.10	3.10	0.10
Neighbour avg	5.10	1.60	0.10	0.10	—
Smoothed	1.40	4.19	0.10	2.32	0.10
$E_N[i]$	0.172	0.516	0.012	0.286	0.012

Link 1 gets **17.2%** despite zero observations $\Rightarrow$ topology propagates evidence from its busy neighbour

Extending to $2N$: The E_{2N} Vector

Teleportation injects vehicles only into the **Up (active)** phase:

$$E_{2N} = \begin{pmatrix} E_N \\ \mathbf{0}_N \end{pmatrix} \in \mathbb{R}^{2N}, \quad \|E_{2N}\|_1 = 1$$

Physical meaning:

- New vehicles are “born” actively driving on a road
- They naturally transition to the Down phase over time
- The ground truth determines *where* they appear; the transition matrix determines *how* they move

Combining Topology + Ground Truth

The Power Iteration

$$\mathbf{v}^{(k+1)} = \underbrace{d \cdot M_{2N}^T \cdot \mathbf{v}^{(k)}}_{\text{topology-driven flow (80\%)}} + \underbrace{(1-d) \cdot E_{2N}}_{\text{ground-truth injection (20\%)}}$$

$$d = 0.80$$

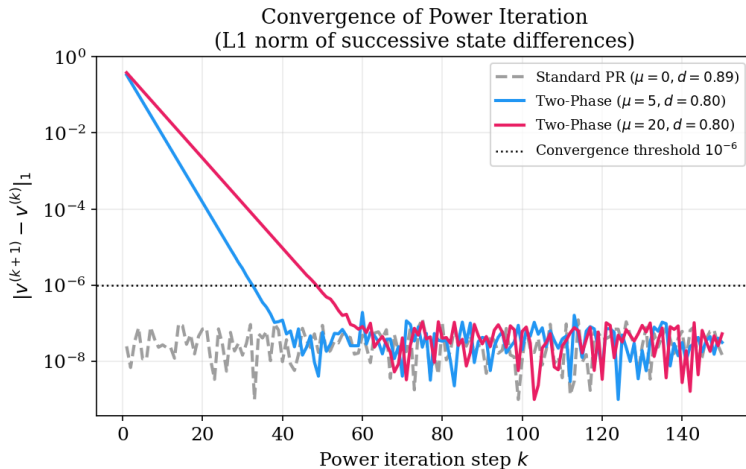
80% of mass follows the
transition matrix
(roads, travel times, phases)

$$1 - d = 0.20$$

20% of mass is re-injected
from **ground-truth GPS**
(observed traffic pattern)

Converges when $\|\mathbf{v}^{(k+1)} - \mathbf{v}^{(k)}\|_1 < 10^{-6}$

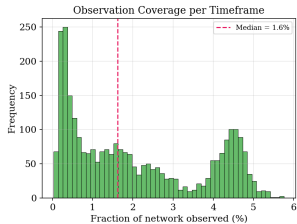
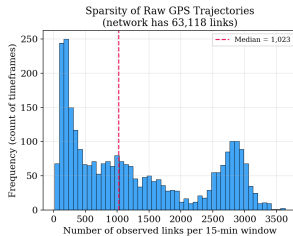
Convergence



Typically converges in ~ 40 – 60 iterations

Why Smooth? — The Sparsity Problem

- Most links have $c_i = 0$
- Creates:
 - MRE singularities
 - Markov chain dead zones
 - Biased teleportation
- Without smoothing:
MRE = 7,275
- With smoothing:
MRE = 0.19



Graph-Diffusion as a Low-Pass Filter

$$\mathbf{c}_{\text{smooth}} = \underbrace{(1 - \gamma)}_{\text{keep own}} \cdot \tilde{\mathbf{c}} + \underbrace{\gamma}_{\text{borrow from neighbours}} \cdot P_{\text{adj}} \tilde{\mathbf{c}}$$

Spectral view:

Eigenvalue of the filter:

$$h(\lambda) = (1 - \gamma) + \gamma\lambda$$

$$h(1) = 1 \text{ (preserves mean)}$$

$$h(-1) = 1 - 2\gamma \text{ (attenuates noise)}$$

Mass conservation:

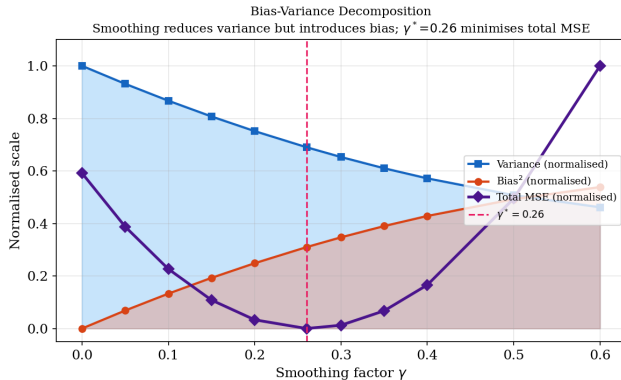
$$\sum_i c_{\text{smooth},i} = \sum_i \tilde{c}_i$$

Smoothing **redistributes**
mass but never creates
or destroys it

The Bias-Variance Tradeoff

$$\text{MSE} = \underbrace{\text{Bias}^2}_{\substack{\uparrow \text{ with } \gamma}} + \underbrace{\text{Variance}}_{\substack{\downarrow \text{ with } \gamma}}$$

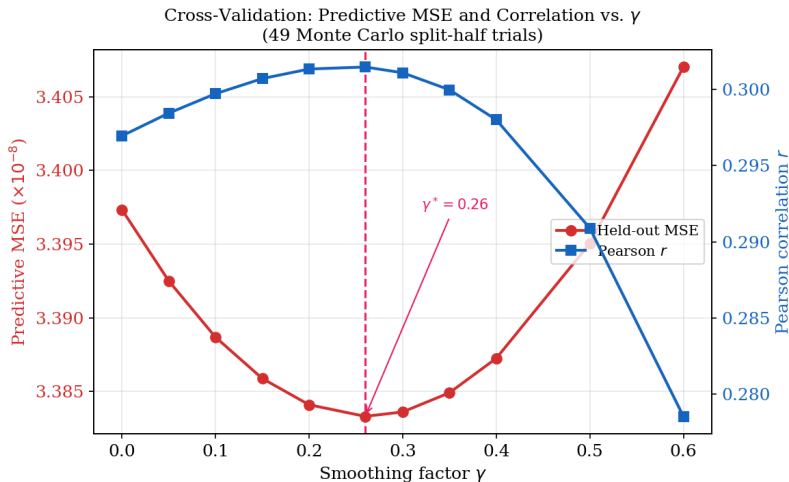
- Too little smoothing → **noisy**
- Too much smoothing → **blurred**
- U-shaped MSE curve
- Minimum **inside** the range



Cross-Validation Protocol (49 Trials)

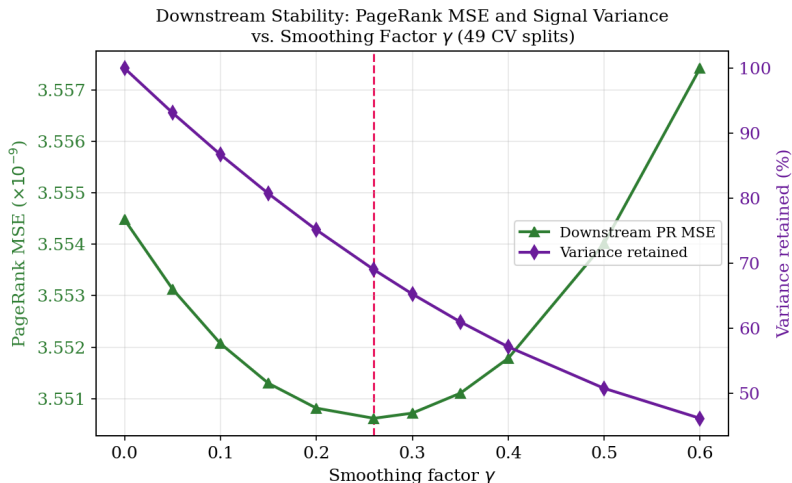
- ① Select 49 timeframe slots (fixed seed = 42)
- ② **Split** GPS data: Set A (train) / Set B (test)
- ③ Smooth Set A with candidate γ
- ④ **Never** smooth Set B (oracle ground truth)
- ⑤ Measure:
 - **Predictive MSE**: smoothed A vs. raw B
 - **PageRank MSE**: PageRank(A) vs. PageRank(B)
 - **Pearson r** : correlation A vs. B
- ⑥ Average over all 49 trials

CV Results: MSE & Correlation



Both metrics optimised at $\gamma^* = 0.26$

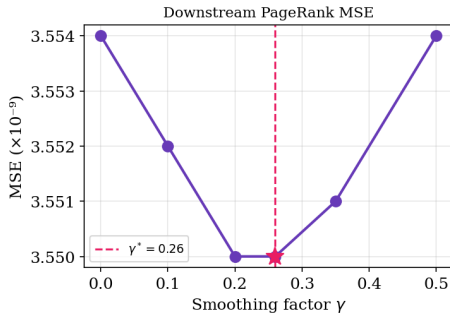
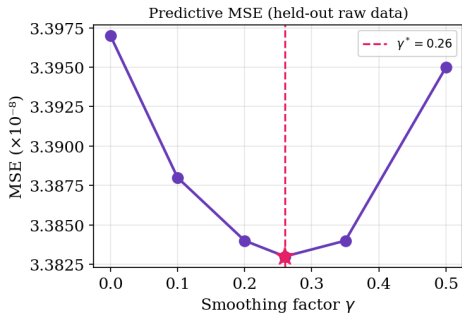
CV Results: Downstream PageRank MSE



PageRank MSE minimum **also** at $\gamma \approx 0.26 \Rightarrow$ input-optimal = output-optimal

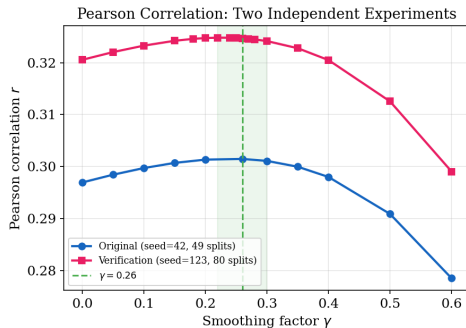
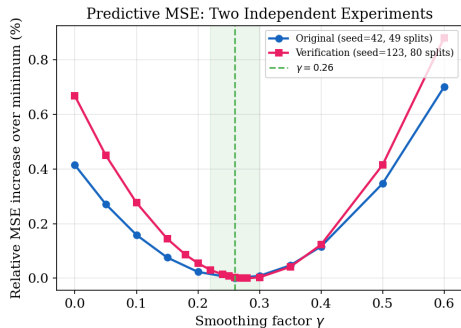
Direct vs. Downstream MSE

Cross-Validation: Optimal Smoothing Parameter ($\gamma^* = 0.26$)



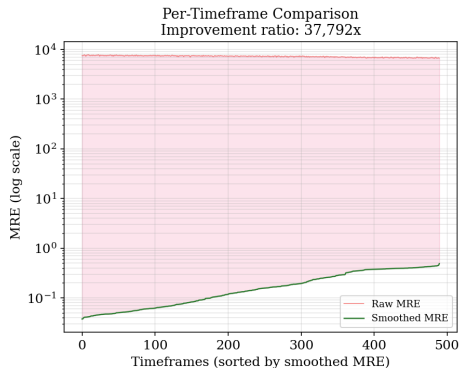
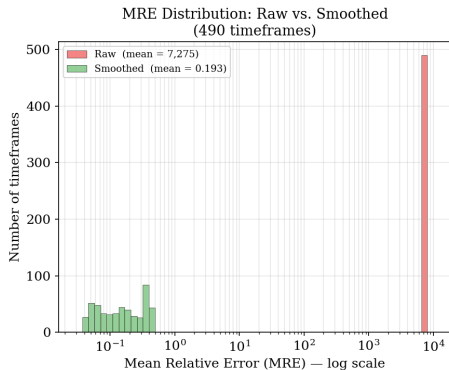
Independent Verification (Seed = 123, 80 Splits)

Independent Verification of Optimal γ^*
(green band = optimal plateau $\gamma \in [0.22, 0.30]$)



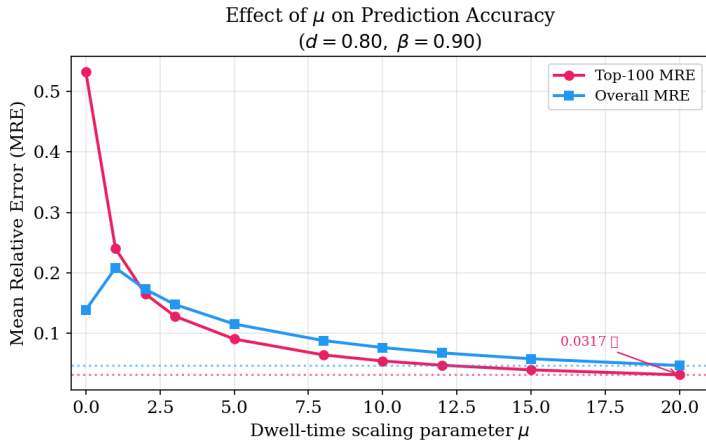
Broad plateau at $\gamma \in [0.22, 0.30]$; $\gamma = 0.26$ is the **centre of the plateau**.
MSE varies $< 0.04\%$ within this band.

Impact of Smoothing: 490-Timeframe Comparison



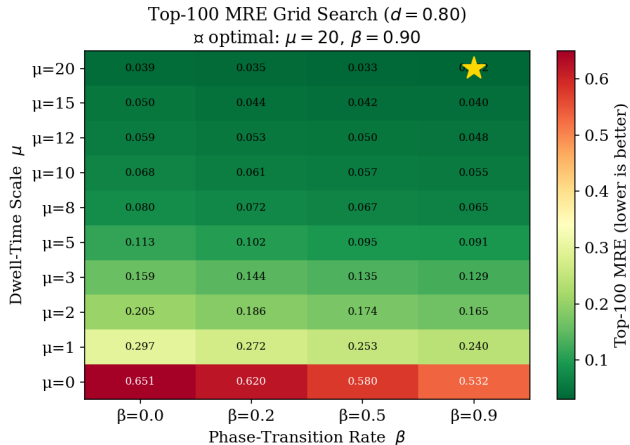
Pipeline	Avg MRE	Improvement
Raw ($\gamma = 0$)	7,275	—
Smoothed ($\gamma = 0.26$)	0.19	$\times 37,800$

Effect of μ (Dwell-Time Scale)



MRE drops monotonically as μ increases. Optimal: $\mu = 20$

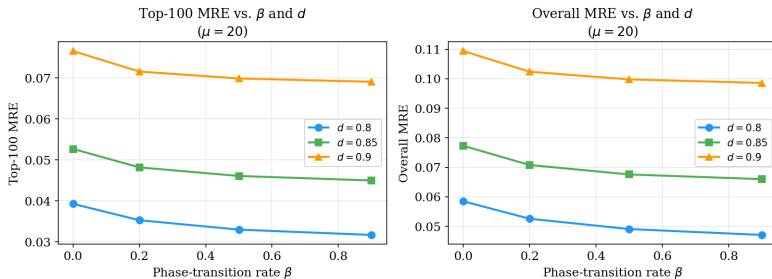
MRE Heatmap: μ vs β



Best combination: $\mu = 20, \beta = 0.90, d = 0.80$

Damping Sensitivity

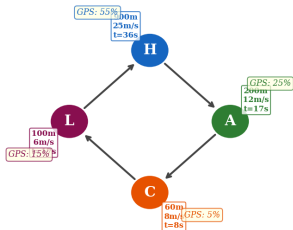
Sensitivity Analysis: Damping Factor d and Phase Rate β



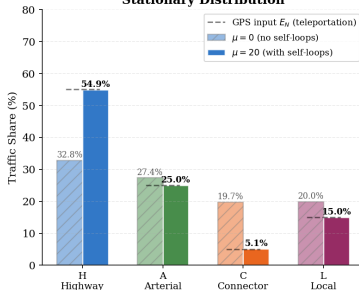
$d = 0.80$ balances topology-driven flow with ground-truth injection

Worked Example: 4-Link Toy Network

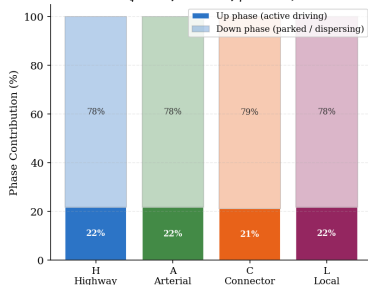
Toy Network Topology



Effect of Self-Loops on Stationary Distribution



Up/Down Phase Decomposition
($\mu = 20, d = 0.80, \beta = 0.90$)



	Highway	Arterial	Connector	Local
GPS input E_N	55%	25%	5%	15%
$\mu = 0$ output	32.8%	27.4%	19.7%	20.0%
$\mu = 20$ output	54.9%	25.0%	5.1%	15.0%

Self-loops **anchor mass** to where vehicles physically spend time

Final Results (49 Timeframes, Seed = 42)

Metric	Value
Top-100 MRE	0.0317
Overall MRE	0.0471

Symbol	Name	Value	Role
γ	Smoothing	0.26	Evidence propagation
μ	Dwell-time	20	Self-loop weight
β	Phase-drop	0.90	Active $\rightarrow$ dispersal
d	Damping	0.80	Topology vs. ground truth

Key Takeaways

- 1 **Transition matrix** M_{2N} encodes road physics:
topology + lengths + speeds $\rightarrow$ two-phase block structure with self-loops
- 2 **Teleportation vector** E_{2N} encodes ground truth:
GPS counts $\rightarrow$ Laplace $\rightarrow$ graph-diffusion smoothing $\rightarrow$ normalise
- 3 **Smoothing** cross-validated at $\gamma^* = 0.26$:
U-shaped MSE, input-optimal = output-optimal
- 4 **Final model**: Top-100 MRE = **0.0317**, Overall MRE = **0.0471**
on 49 random 15-minute timeframes from September 2018

Thank You

Questions?